

S3 Appendix: Deriving the marginal distribution of the observed tract length under two alternative settings

We first consider the case in which N is distributed as a sum of two independent and identically distributed geometric random variables each with mean $\phi/2$. We have,

$$P(N = n) = (n - 1) \left(1 - \frac{2}{\phi}\right)^{n-2} \left(\frac{2}{\phi}\right)^2.$$

Letting $\gamma = \frac{2}{\phi}$,

$$P(L = l) = \sum_{n=l}^{\infty} P(L = l|N = n)P(N = n)$$

$$= \begin{cases} \frac{\gamma^2(1-\psi)^2}{(\gamma + \psi - \gamma\psi)^2} & \text{if } l = 0 \\ \frac{2\gamma^2\psi(1-\psi)}{(\gamma + \psi - \gamma\psi)^3} & \text{if } l = 1 \\ \frac{\gamma^2(1-\gamma)^{l-2}\psi^2[(l-3)(\gamma + \psi - \gamma\psi) + 2]}{(\gamma + \psi - \gamma\psi)^3} & \text{if } l \geq 2. \end{cases}$$

Then,

$$P(2 \leq L \leq M) = \sum_{l=2}^M \frac{\gamma^2(1-\gamma)^{l-2}\psi^2[(l-3)(\gamma + \psi - \gamma\psi) + 2]}{(\gamma + \psi - \gamma\psi)^3}$$

$$= \frac{(\gamma + \psi - \gamma\psi)\psi^2[(3-M)\gamma(1-\gamma)^{M-1} - (1-\gamma)^{M-1} - 2\gamma + 1] + 2\gamma\psi^2[1 - (1-\gamma)^{M-1}]}{(\gamma + \psi - \gamma\psi)^3}.$$

Then,

$$P(L = l|2 \leq L \leq M) = \frac{P(L = l)}{P(2 \leq L \leq M)}$$

$$= \frac{(\gamma + \psi - \gamma\psi)(l-3)\gamma^2(1-\gamma)^{l-2} + 2\gamma^2(1-\gamma)^{l-2}}{(\gamma + \psi - \gamma\psi)[(3-M)\gamma(1-\gamma)^{M-1} - (1-\gamma)^{M-1} - 2\gamma + 1] + 2\gamma[1 - (1-\gamma)^{M-1}]}.$$

Similarly to the case where N is geometric, we index our random variable L using j so that L_j represents the random variable corresponding to the observed tract length for detected tract j in our dataset. This time, we also index ψ using j so that an allele conversion happens with probability ψ_j at every position within the j th detected tract (the estimation of ψ_j is described in the section, Estimating the allele conversion probability for each detected tract). We have,

$$P(L_j = l_j | 2 \leq L_j \leq M) = \frac{(\gamma + \psi_j - \gamma\psi_j)(l_j - 3)\gamma^2(1 - \gamma)^{l_j-2} + 2\gamma^2(1 - \gamma)^{l_j-2}}{(\gamma + \psi_j - \gamma\psi_j)[(3 - M)\gamma(1 - \gamma)^{M-1} - (1 - \gamma)^{M-1} - 2\gamma + 1] + 2\gamma[1 - (1 - \gamma)^{M-1}]}.$$

We next consider the case where N is distributed as a mixture of two geometric components. We let the two geometric means be ϕ_1 and ϕ_2 , and let w_1 represent the mixing proportion of the first component. We have,

$$P(N = n) = w_1 \left(1 - \frac{1}{\phi_1}\right)^{n-1} \frac{1}{\phi_1} + (1 - w_1) \left(1 - \frac{1}{\phi_2}\right)^{n-1} \frac{1}{\phi_2}.$$

Letting $\lambda_1 = 1/\phi_1$ and $\lambda_2 = 1/\phi_2$,

$$P(L = \ell) = \sum_{n=\ell}^{\infty} P(L = \ell | N = n) P(N = n)$$

$$= \begin{cases} \frac{w_1 \lambda_1 (1 - \psi)}{\lambda_1 + \psi - \lambda_1 \psi} + \frac{(1 - w_1) \lambda_2 (1 - \psi)}{\lambda_2 + \psi - \lambda_2 \psi} & \text{if } \ell = 0 \\ \frac{w_1 \lambda_1 \psi}{(\lambda_1 + \psi - \lambda_1 \psi)^2} + \frac{(1 - w_1) \lambda_2 \psi}{(\lambda_2 + \psi - \lambda_2 \psi)^2} & \text{if } \ell = 1 \\ \frac{w_1 \lambda_1 (1 - \lambda_1)^{\ell-1} \psi^2}{(\lambda_1 + \psi - \lambda_1 \psi)^2} + \frac{(1 - w_1) \lambda_2 (1 - \lambda_2)^{\ell-1} \psi^2}{(\lambda_2 + \psi - \lambda_2 \psi)^2} & \text{if } \ell \geq 2. \end{cases}$$

Then,

$$\begin{aligned}
P(2 \leq L \leq M) &= \sum_{l=2}^M \left[\frac{w_1 \lambda_1 (1 - \lambda_1)^{l-1} \psi^2}{(\lambda_1 + \psi - \lambda_1 \psi)^2} + \frac{(1 - w_1) \lambda_2 (1 - \lambda_2)^{l-1} \psi^2}{(\lambda_2 + \psi - \lambda_2 \psi)^2} \right] \\
&= \frac{w_1 \psi^2 [(1 - \lambda_1) - (1 - \lambda_1)^M]}{(\lambda_1 + \psi - \lambda_1 \psi)^2} + \frac{(1 - w_1) \psi^2 [(1 - \lambda_2) - (1 - \lambda_2)^M]}{(\lambda_2 + \psi - \lambda_2 \psi)^2}.
\end{aligned}$$

Then,

$$\begin{aligned}
P(L = \ell | 2 \leq L \leq M) &= \frac{P(L = \ell)}{P(2 \leq L \leq M)} \\
&= \frac{\frac{w_1 \lambda_1 (1 - \lambda_1)^{\ell-1} \psi^2}{(\lambda_1 + \psi - \lambda_1 \psi)^2} + \frac{(1 - w_1) \lambda_2 (1 - \lambda_2)^{\ell-1} \psi^2}{(\lambda_2 + \psi - \lambda_2 \psi)^2}}{\frac{w_1 \psi^2 [(1 - \lambda_1) - (1 - \lambda_1)^M]}{(\lambda_1 + \psi - \lambda_1 \psi)^2} + \frac{(1 - w_1) \psi^2 [(1 - \lambda_2) - (1 - \lambda_2)^M]}{(\lambda_2 + \psi - \lambda_2 \psi)^2}}.
\end{aligned}$$

Again using j to index detected tracts,

$$P(L_j = \ell_j | 2 \leq L_j \leq M) = \frac{\frac{w_1 \lambda_1 (1 - \lambda_1)^{\ell_j-1} \psi_j^2}{(\lambda_1 + \psi_j - \lambda_1 \psi_j)^2} + \frac{(1 - w_1) \lambda_2 (1 - \lambda_2)^{\ell_j-1} \psi_j^2}{(\lambda_2 + \psi_j - \lambda_2 \psi_j)^2}}{\frac{w_1 \psi_j^2 [(1 - \lambda_1) - (1 - \lambda_1)^M]}{(\lambda_1 + \psi_j - \lambda_1 \psi_j)^2} + \frac{(1 - w_1) \psi_j^2 [(1 - \lambda_2) - (1 - \lambda_2)^M]}{(\lambda_2 + \psi_j - \lambda_2 \psi_j)^2}}.$$

In practice, we plug in $M = 1500$ because we exclude all observed tract lengths longer than 1500 bp detected from the UK Biobank whole autosome data.